

Program 1

Design and implement C/C++ Program to find Minimum Cost Spanning Tree of a given connected undirected graph using Kruskal's algorithm.

```
#include<stdio.h>
int min=0,cost[10][10],parent[10],i,j,x,y,n;
void main()
{
    int count=0,tot=0,flag=0;
    void find_min();
    int check_cycle();
    printf("enter the no of vertices\n");
    scanf("%d",&n);
    printf("enter the matrix\n");
    printf("enter 999 for self loops and no edge\n");
    for(i=1;i<=n;i++)
    for(j=1;j<=n;j++)
    {
        scanf("%d",&cost[i][j]);
        parent[j]=0;
    }
    while(count!=n-1 &&min!=999)
    {
        find_min();
        flag=check_cycle(x,y);
        if(flag==1)
        {
            printf("%d-->%d=%d\n",x,y,cost[x][y]);
            count++;
            tot+=cost[x][y];
        }
        cost[x][y]=cost[y][x]=999;
    }
    printf("the total cost=%d",tot);
}

void find_min()
{
    min=999;
    for(i=1;i<=n;i++)
    for(j=1;j<=n;j++)
    if(cost[i][j]<min)
    {
        min=cost[i][j];
        x=i;
        y=j;
    }
}
```

```

int check_cycle(int x,int y)
{
    if((parent[x]==parent[y])&&(parent[x]!=0))
        return 0;
    else if(parent[x]==0&&parent[y]==0)
        parent[x]=parent[y]=x;
    else if(parent[x]==0)
        parent[y]=parent[x];
    else if(parent[y]==0)
        parent[y]=parent[x];
    else if(parent[x]!=parent[y])
        parent[y]=parent[x];
    return 1;
}

```

Sample Input and Output:

enter the no of vertices

4

enter the matrix

enter 999 for self loops and no edge

999 1 5 2

1 999 999 999

5 999 999 3

2 999 3 999

1-->2=1

1-->4=2

3-->4=3

the total cost=6